University of Cologne

Master Thesis in Economics

Your great thesis title

Your Name

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Submitted for Examination in the Study Programme *Economics (M.Sc.)*
at the *Faculty of Management, Economics and Social Sciences*

Cologne, October 1, 2020

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1. Introduction

Here goes your introduction. This document can be used as a template (in this case you want to delete all my text from here to the `\bibliographystyle{chicago}` comand) but I will also give some examples of how to cite and reference, and how to use mathematics in LaTeX. Ideally, you should look at the .pdf and the .tex file at the same time to see what the code generates.

1.1. Literature Review

The literature review goes here. Let me give some examples how to cite in LaTeX using BibTeX: I cite Stigler (1980) and Posner (1981) as examples. A page in a book is cited as Solove (2010, p. 34). Note that all the cited articles, books etc. have to have an entry in the .bib file where you define the label used for citing and store all the information necessary for creating the reference list at the end of the document. You can usually copy and paste the entries for the .bib file from Google Scholar or JSTOR (and otherwise you do it by hand following Martin Osborne's guide).

The advantage of BibTeX is twofold: First, you spell the author names and the publication years correctly. Second, the reference list is automatically generated and formatted nicely.

Let me also comment on some common problems. First, how to cite a website? I suggest to cite websites as in Wikimedia Foundation (2003) and articles on news websites or blogs as in Ernst (2020). Second, capitalization in the reference list: The here chosen bibstyle will put all but the first letter of the title (of a journal article) in lower case. This should be avoided for names etc. To do so put capital letters in between curly brackets in the .bib file, e.g. "On {N}ash equilibrium" instead of "On Nash equilbirium". You can see an example for this in the BibTeX entry Ernst (2020).

2. Mathematics

Mathematics within a line of text is put between $ marks. For example, `$\frac{3}{4}$` yields the fraction $\frac{3}{4}$ or $\lim_{x\rightarrow \infty}e^{-x}=0$ is generated by `$\lim_{x\rightarrow \infty}e^{-x}=0$`. Subindeces as in x_1 are written as `$x_1$`. Here is how you use math environments:

Definition 1 (Fixed Point)**.** *Let $f : D \to R$ where $D \subseteq R$. If $f(x) = x$, then x is called a* fixed point *of f.*

You reference the definition as "definition 1" (that is `definition \ref{def:fp}` in the source file where `def:fp` is the label of the definition above). Similarly, you can reference sections as `section \ref{sec:intro}` which yields "section 1". Also subsection 1.1 can be referenced using `subsection \ref{sec:literature}`. The advantage of such references

over writing "section 1.1" directly is that your references will automatically adjust if you change something in the structure of your document; for example, if you insert another section. (Note that you have to compile LaTeX twice before it gets a new reference properly formatted.)

Note that you can generate numbered equations that can be similarly referenced as follows:

$$f(x) = x. \tag{1}$$

This equation can be referenced as equation 1.

3. Results

A mathematical result, like a proposition, is written in a similar environment as a definition.

Proposition 1. *Let $f : [0,1] \to [0,1]$ be an increasing function. Then, f has a fixed point.*

4. Figures and Tables

You can include a graphic with the command `\includegraphics{filename}` (recommended file formats are png, eps or pdf). Usually it is a good idea to place the graphic in a figure environment as here:

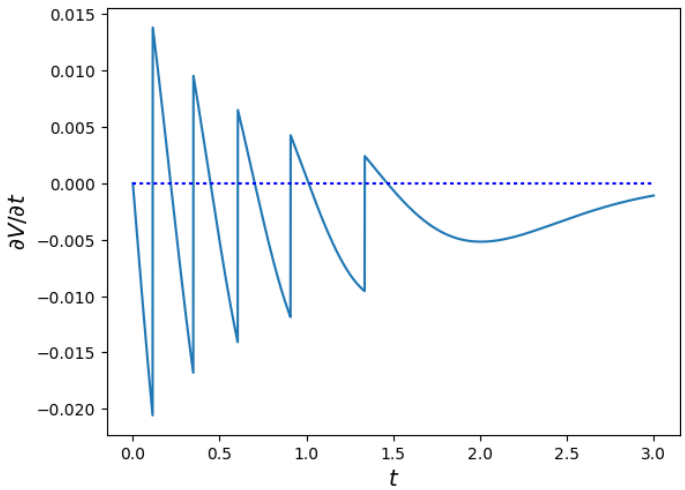

Figure 1: A nice figure

We can reference the figure as figure 1 using its label.

Sometimes you need to have a nicely formatted table. I give a simple example; see `https://en.wikibooks.org/wiki/LaTeX/Tables` for more on this. In case you want to present a regression output in a table, please note that there are packages producing LaTeXtables for the most common statistics programmes.

Country	Capital	Population	Source
China	Beijing	1,404,556,640	National population clock
Japan	Tokio	125,930,000	Monthly provisional estimate
Germany	Berlin	83,157,201	National quarterly estimate

Table 1: Countries and their population

Country	Capital	Population	Source
China	Beijing	1,404,556,640	National population clock
Japan	Tokio	125,930,000	Monthly provisional estimate
Germany	Berlin	83,157,201	National quarterly estimate

Table 2: Countries and their population a bit simpler

5. Conclusion

All scientific writing should be done in LaTeX.

Appendix

A. Proofs

Proof of proposition 1: Let

$$A = \{x : \ x \leq f(x)\}. \tag{2}$$

Note that $0 \in A$ and therefore $A \neq \emptyset$. Let $x^* = \sup A$. Then $f(x^*) \geq x^*$ by the definition of A, see equation (2), and because f is increasing.

Also $f(x^*) \leq x^*$: if not, i.e. $f(x^*) > x^*$, then $x^* + \varepsilon < f(x^* + \varepsilon)$ for some $x^* + \varepsilon \in (x^*, f(x^*)] \cap [0, 1]$ because f is increasing which contradicts the definition of x^*. Hence, $x^* = f(x^*)$. $\square$

B. Second Appendix

Another section in the appendix.

References

Ernst, A. (2020). Was haben Sanktionen gegen Lukaschenko mit der Türkei zu tun? Nichts, doch Zypern blockiert mit der Verknüpfung die EU-Aussenpolitik. https://www.nzz.ch/international/eu-sanktionen-gegen-weissrussland-zypern-blockiert-ld.1577839. Accessed: 2020-10-01.

Posner, R. A. (1981). The economics of privacy. *American Economic Review 71*(2), 405–409.

Solove, D. J. (2010). *Understanding Privacy*. Cambridge, MA: Harvard University Press.

Stigler, G. J. (1980). An introduction to privacy in economics and politics. *Journal of Legal Studies 9*(4), 623–644.

Wikimedia Foundation (2003). LaTeX wikibook. https://en.wikibooks.org/wiki/LaTeX. Accessed: 2020-09-30.